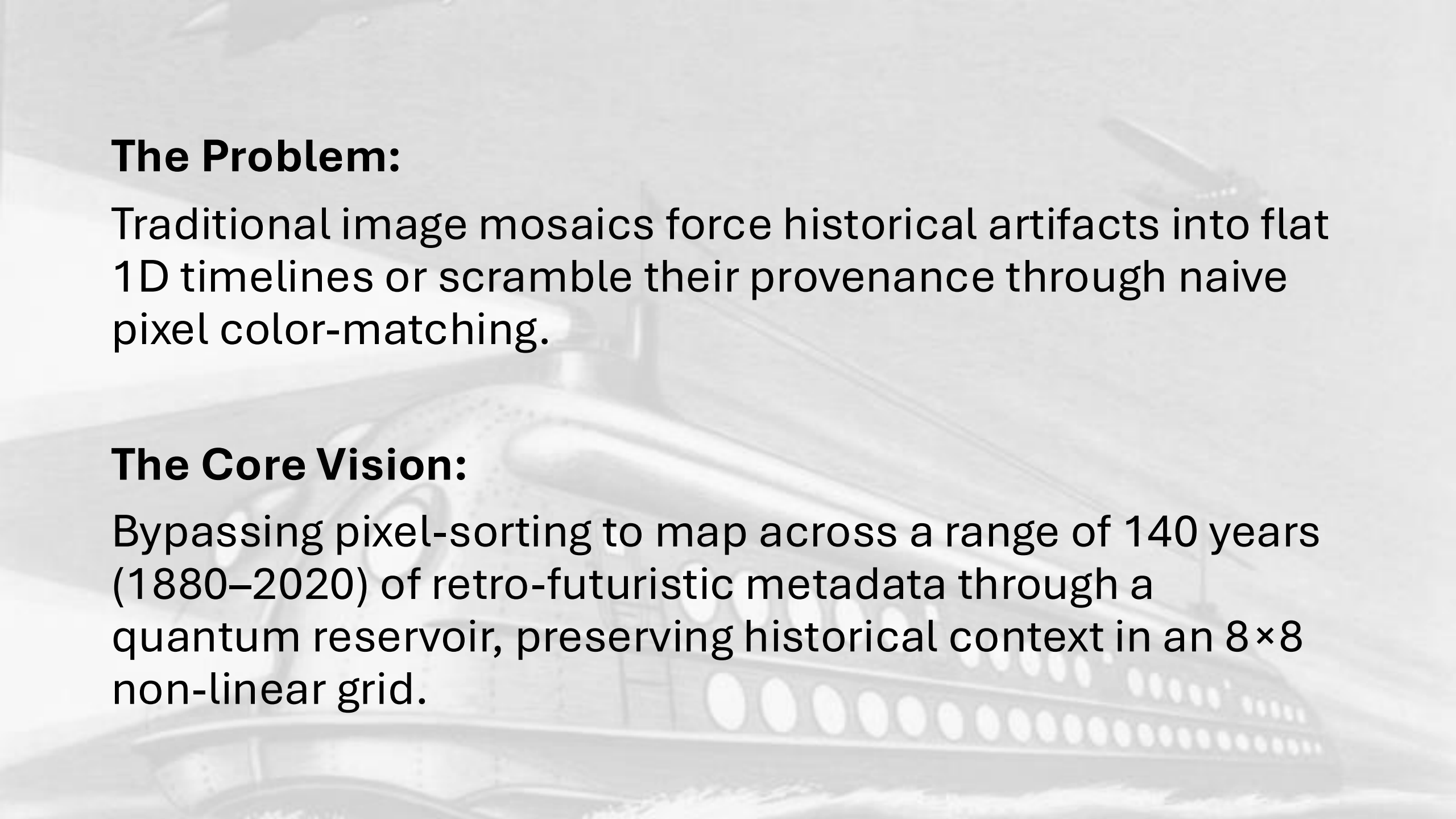


QRC-Mosaic: Mapping Retro-Futuristic Trajectories via Quantum Reservoir Computing

Moth_Hack Quantum Computing Hackathon 2026 Submission





The Problem:

Traditional image mosaics force historical artifacts into flat 1D timelines or scramble their provenance through naive pixel color-matching.

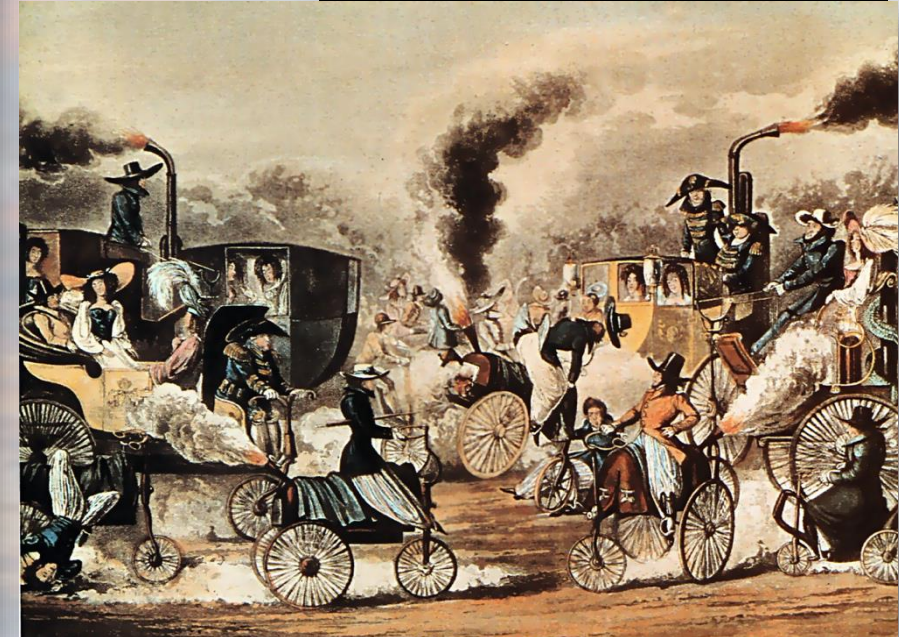
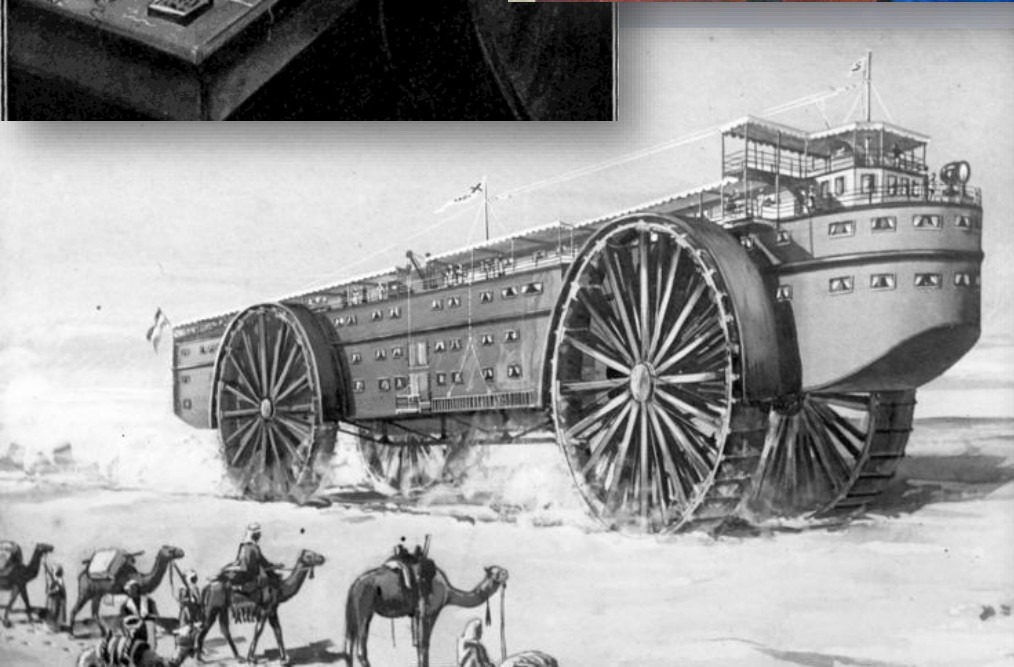
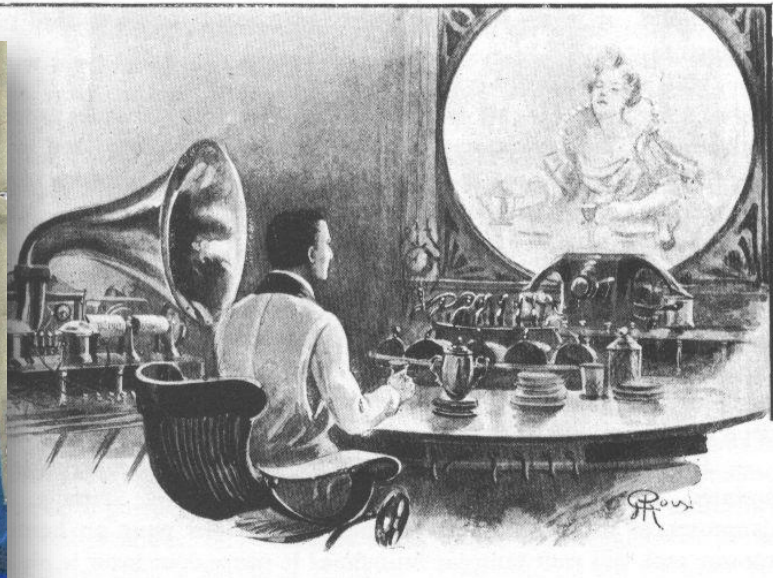
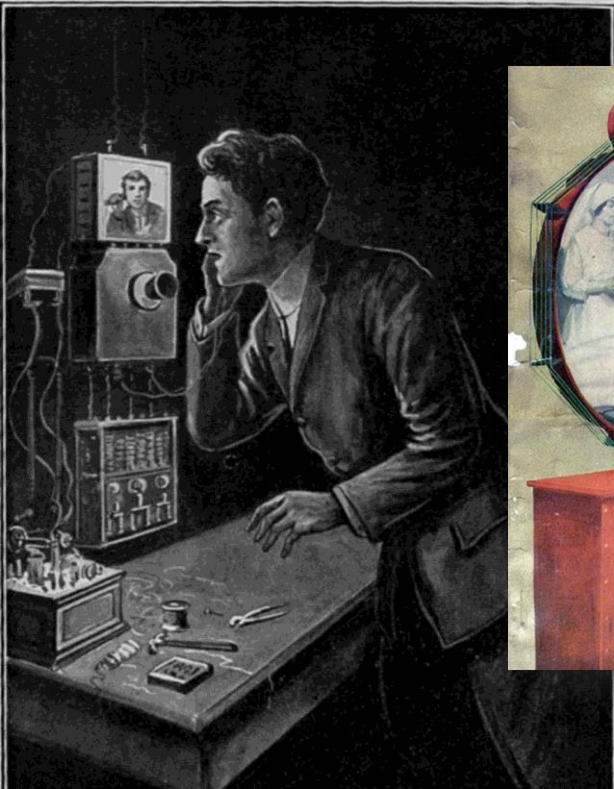
The Core Vision:

Bypassing pixel-sorting to map across a range of 140 years (1880–2020) of retro-futuristic metadata through a quantum reservoir, preserving historical context in an 8×8 non-linear grid.

Quantum-VLM Hybrid Pipeline: Architecture & MOTH Engines

The logo for MOTH, featuring a stylized 'M' symbol followed by the word 'MOTH' in a bold, sans-serif font, all contained within a black rectangular box.

- **Classical Preprocessing of Images** (for benchmarking): MobileNetV3 visual embeddings & CIELAB colour feature extraction.
- **Multimodal Vision-Language Annotation of images:**
Open-weights VLP **Qwen2.5-VL-7B-Instruct** inference (via Hugging Face) extracting 13 semantic category layers across the individual image tiles.
- **MOTH Atlas Integration** (finding non-linear similarities through the ages):
Chronological dates processed through MOTH Atlas engines
qrc-train-v2 & **qrc-gen-v2** via ~65,000+ quantum measurement shots per epoch across the 64-tile grid.



VLP: Qwen2.5-VL-7B-Instruct Inference

13 layers categorise structured metadata such as:

- **Primary Objects** (e.g., rockets, monorails, domes, control panels)
- **Creator Lineage** (e.g., NASA/JPL, archival illustrator)
- **Geographical Origin / Hub** (e.g. North America, Europe, East Asia)
- **Tech-Centric Thematic Era** (e.g. Steam Age, Space Race, Atomic Age, Cybernetic Era)
- **Architectural / Environmental Setting** (e.g. orbital station, alien landscape, futuristic city)
- **Physical Materiality** (e.g. oil painting, travel poster, technical diagram)
- **Human Activity / Subject Matter** (e.g., exploration, labour, observation, leisure)
- **Atmospheric Mood / Tone** (e.g. utopian optimism, industrial dystopia)
- **Visual Density & Line Profile**
- **Luminance & Contrast Distribution**
- **Dominant Spatial Composition**
- **Colour Profile Cues** (high-level overall palette, tonal balance and visual mood)
- **Fictional / Speculative Technology Type**

QRC-Mosaic: 4-Notebook Pipeline Architecture

End-to-End Hybrid Quantum-Classical & Multimodal VLM Workflow

STEP 01

Feature Extraction & Tagging

01_qrc_feature_extraction...

- CIELAB Color Space (L*a*b*)
- MobileNetV3 Embeddings
- 512x512 Crop Scaling

► Output: image_metadata.csv

STEP 02

VLM Semantic Annotation

02_vlm_semantic...

- Qwen2.5-VL-7B Model
- 13 Semantic Categories
- Provenance & Era Tags

► Output: descriptions.json

STEP 03

Quantum Reservoir Permutation

00_qrc_mosaic_mapping...

- MOTH Atlas Engines
- qrc-train-v2 & qrc-gen-v2
- **5 Qubits** (4 Feature + 1 Ancilla)
- **~65k+ Shots/Epoch** (192k Prod)

► Output: Diagnostic Lenses

STEP 04

Dual-Eye Animation & Export

03_aux_dynamic_dual_eye...

- Master Anchor Overlay
- Eyelid Gating & Pupil Track
- Metadata View Crossfade

► Output: animated_eyes.gif

Performance Benchmark & The 21.8% Gap

Key Quantitative Results

Metric	Classical Hungarian	QRC Permutation
Color Fidelity (ΔE)	1.8563 (Ideal Color)	2.3747 (+21.8% Gap)
Sorting Intent	Surface RGB Matching	High-Dim Metadata Trajectory

Key Takeaway:

Despite being completely blind to image colours, pure quantum temporal ordering landed within 21.8% of a classical algorithm forced strictly to match pixels.



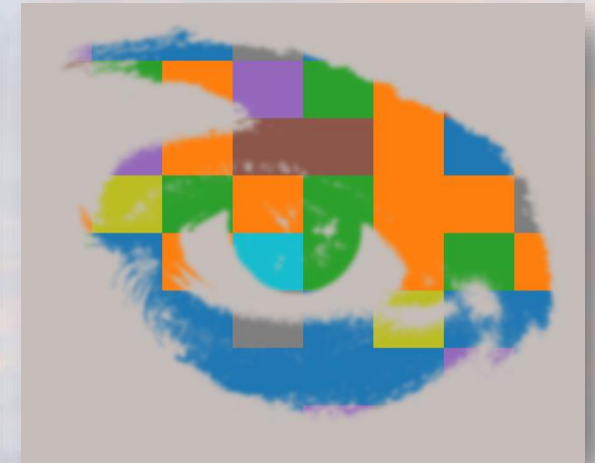
Topological Diagnostic Lenses: Post-Hoc Validation

- **Statistical Validation:**
Post-hoc spatial analysis confirmed statistically significant clustering ($Z=+2.07$)
- **Discovered Topologies:**
Automatically formed spatial clusters across Creator Lineages (e.g., NASA/JPL material), Geographical Origins, and Tech Eras without explicit spatial instructions.

Year Overlay



Semantic Overlay



Creator Overlay



Geography Overlay



Final Mosaic Calculation and Softening



Classical Transitioning & Animation Applied

